

USB as the Power Delivery Source

Power Delivery USB A 2.0/3.0

+5V	_____	500mA (USB 2.0)	→	2.5W	→ wireless mouse
	_____	900mA (USB 3.0)	→	4.5W	→ external hard drive
	_____	up to 1.5A	→	7.5W	→ smartphone (slow charging)

- To use 1.5A, device must have a special resistor on the data lines to signal "I need more power" (USB Battery Charging spec).
- However, it's not a good idea to draw this level of current due to voltage drop ($V = IR$) and potential damage to the port.

Warning with USB A ports/cables:



- USB A ports can provide power, but they have limitations in terms of current and voltage, which can lead to issues when trying to power high-demand devices.
- For higher power devices, prefer USB-C (ideally with Power Delivery) rather than pulling high current from a USB-A port.

Qualcomm Quick Charge (QC)

Quick Charge (QC) is Qualcomm's proprietary fast-charging protocol.

- It runs over USB-A or USB-C and uses the **D+ / D- data lines** (not CC) to negotiate voltage.
- QC was designed to work with older USB-A devices, but it is a separate ecosystem from USB PD.
- Many modern chargers support both QC and PD simultaneously.

USB A can use QC

QC uses the D+/D- data pins.

- This allows QC to be used with older USB-A devices, but it also means QC is a separate ecosystem from USB PD.
- QC 1.0 can support up to 10W (5V/2A)
- QC 2.0 up to 18W (20V/0.9A) with 5/9/12/20 with 5/9/12/20V
- QC 3.0 also up to 18W (from 3.6V) but with finer voltage control.

QC negotiation (simplified):

Device: sets $D+ = 0.6V$, $D- = 0V$ → "I support Quick Charge"
Charger: detects signal → "OK, I'll raise voltage"
Device: sets $D+ = 3.3V$, $D- = 0.6V$ → "Give me 9V"
Charger: switches VBUS to 9V → faster charging begins



- To use QC, both the charger and device must support it.
- If either side doesn't, it falls back to standard 5V charging.

The better solution to draw more power is to use USB-C with Power Delivery (PD) protocol:

Negotiation conversation:

Device: "I can accept 5V, 9V, 15V, or 20V"

Host: "I can provide up to 20V at 5A"

Device: "Give me 20V at 3A please"

Host: "OK, switching to 20V/3A" → 60W laptop charging

Real devices:

- Raspberry Pi 4: needs $5V/3A = 15W$
- Raspberry Pi 5: needs $5V/5A = 25W$
- MacBook Pro: negotiates $20V/4.5A = 90W$
- USB-powered monitor: $20V/3.25A = 65W$

Normal USB C cables cannot handle more than 3A, so USB-C cables must be designed for higher current and have proper markings (e.g., "5A" or "100W") with e-marker chip to ensure safe operation.

e-Marker Chips in USB-C Cables

- e-Marker (electronic marker) chips are embedded in USB-C cables to communicate the cable's capabilities to the host and device.



- They provide information such as:
 - Maximum current rating (e.g., 3A or 5A)
 - Supported power levels (e.g., up to 100W)
 - Data transfer capabilities (e.g., USB 2.0, USB 3.1, Thunderbolt)

So, the USB-C cables with an e-Marker (100W) & high speed support, such as USB4 (20–40+ Gbps), are much more expensive than basic USB-C cables without e-Marker.

USB PD Versions — How the Standard Evolved

Version	Year	Max Power	Key Addition
USB PD 1.0	2012	100W	First PD spec; 5V/9V/15V/20V profiles
USB PD 2.0	2016	100W	Merged into USB-C spec; PPS added
USB PD 3.0	2019	100W	PPS (Programmable Power Supply)
USB PD 3.1	2021	240W	Extended Power Range (28V/36V/48V)

PPS (Programmable Power Supply):

- Instead of fixed voltage steps, PPS allows fine-grained voltage adjustment
- Example: charge at exactly 8.76V / 2.25A instead of jumping between 9V and 5V
- Used in modern fast-charging smartphones (Samsung, Xiaomi)

QC Versions Compared

Version	Year	Max Power	Voltages Available	Connector	Key Feature
QC 1.0	2013	10W	5V only	USB-A	2× faster than standard USB; fixed 5V
QC 2.0	2014	18W	5V / 9V / 12V / 20V	USB-A / USB-C	First multi-voltage QC; 4 fixed steps
QC 3.0	2016	18W	3.6V – 20V (200mV steps)	USB-A / USB-C	INOV: fine-grained voltage like PPS
QC 4.0	2017	27W	3.6V – 20V + PPS	USB-C only	Merges with USB PD 3.0; adds PPS
QC 4+	2018	27W	PPS + dual-charge	USB-C only	Dual-charge path; better thermal control
QC 5.0	2020	100W	3.3V – 20V PPS	USB-C only	70% charge in 15 min; 4× faster than QC 3.0

QC vs USB PD — Key Differences

Feature	Quick Charge (QC)	USB PD
Made by	Qualcomm (proprietary)	USB-IF (open standard)
Negotiation via	D+ / D– data pins	CC pin
Works on USB-A?	Yes (QC 1.0–3.0)	No (USB-C only)
Max power	100W (QC 5.0)	240W (PD 3.1)
Fine voltage control	QC 3.0+ (200mV steps)	PPS (20mV steps, finer)
Compatibility	Needs matching QC charger + device	Broader cross-vendor support
IoT use	Common in Snapdragon-based gateways	Preferred for custom IoT boards

Practical note: Many modern chargers support **both QC 3.0 and USB PD** simultaneously. Devices negotiate whichever protocol they understand.

The CC Pin — The Brain of USB-C Power

The **CC (Configuration Channel)** pins are what make USB-C "smart."

USB-C Connector (simplified):

VBUS	VBUS	GND	GND		
CC1	CC2				
D+	D-	TX1	RX1	TX2	RX2

← power rails
← configuration pins
← data

CC pins serve three roles:

1. **Cable orientation detection** — which way is the plug inserted?
2. **Role negotiation** — who is host (source) vs. device (sink)?
3. **PD protocol** — voltage/current negotiation messages travel over CC

Without CC logic, a USB-C port is just "dumb 5V" like old USB-A.

How PD Negotiation Works Step by Step

- Step 1: Cable plugged in CC resistor detected
 - host knows a device is connected
- Step 2: Host sends Source_Capabilities
 - "I offer: 5V/3A, 9V/3A, 15V/3A, 20V/5A"
- Step 3: Device reads list → sends Request
 - "I want: 20V/3A please (60W)"

- Step 4: Host sends Accept + Power_Supply_Ready (PS_RDY)
 - "OK, I am switching to 20V now"
- Step 5: Voltage transitions → power flows at agreed level

Total time: ~100 milliseconds

If no PD negotiation happens, the default is always **5V / 0.9A** (USB default).

USB-C Cable Types — Not All Are Equal

Cable Type	Max Current	Max Power	E-Marker?	Data Speed
Basic USB-C (56kΩ)	3A	60W	No	USB 2.0
Full-featured USB 3.2	3A	60W	No	10 Gbps
5A rated cable	5A	100W	Yes	USB 3.2
USB4 Gen2	5A	100W	Yes	20 Gbps
USB4 Gen3 / Thunderbolt 4	5A	240W (PD 3.1)	Yes	40 Gbps

Without PD (USB C): $5V \times 3A = 15W$ max, regardless of cable claims. Risk of overheating if you try to draw more.

What Happens When You Use the Wrong Cable

Scenario 1: Cheap cable (no e-Marker) + 100W charger + 100W laptop

- Charger reads: "no e-Marker = max 3A cable"
- Charger limits to 60W
- Laptop charges slowly but safely ✓ (safe but slow)

Scenario 2: Counterfeit cable (claims 5A but no e-Marker)

- Charger sees no e-Marker → limits to 60W ✓
- But cable internal wire is thin → runs hot at 3A x
- Risk of insulation melt, fire x

Scenario 3: Correct 5A e-Marked cable + 100W charger + 100W laptop

- Charger reads e-Marker: "5A cable confirmed"
- Negotiates 20V/5A = 100W
- Fast, safe charging ✓✓

- Buy cables from reputable brands.
- Chargers use the e-Marker to determine 5A capability, but poorly made cables can falsely claim compliance.
- Without PD (USB-C): Maximum is up to $5V \times 3A$ (15W) if the source advertises 3A; otherwise it may be limited to 0.9A or 1.5A.
- Without PD (USB-A): Standard charging is up to $5V \times 1.5A$ (7.5W); higher power requires proprietary fast-charge protocols.

USB-C Roles — Source, Sink, and DRP

USB-C defines three power roles:

Source — provides power (charger, power bank, PC)

Sink — consumes power (phone, laptop, IoT device)

DRP (Dual Role Power) — can be either source or sink

Example: Laptop with DRP port

- Plugged into charger: acts as Sink (receives power)
- Plugged into USB drive: acts as Source (provides power)
- Role determined during CC negotiation at plug-in
- Role determined during CC negotiation at plug-in

For IoT:

- Sensors / embedded devices → always **Sink**
- IoT gateway / hub powering peripherals → **DRP** or **Source**

Setting CC pins

For IoT devices that use USB-C for power, the device will be a **Sink** (power consumer).

CC Resistors

USB-C role is defined electrically on the **CC pins**, not by the connector shape.

- **R_p (pull-up)** → Source
- **R_d (pull-down)** → Sink
- Toggle R_p/R_d → DRP (Dual Role Power)

IoT board (power input):

CC1 → Rd
CC2 → Rd

→ Always Sink

Charger:

CC1 → Rp
CC2 → Rp

→ Always Source

USB-C Power in Practice — Raspberry Pi 5

The Raspberry Pi 5 is a great IoT gateway example:

RPi 5 power requirements:

Idle: $5V / 1A = 5W$

Under load: $5V / 3A = 15W$

With USB periphs: $5V / 5A = 25W$ (recommended PSU)

What happens with different chargers:

5V/1A USB-A charger → RPi 5 boots but throttles CPU x
5V/3A USB-C (no PD) → Works for light use only
5V/5A PD charger → Full performance ✓✓
Official RPi 5 PSU → 27W PD, powers everything ✓✓✓

How to check on RPi 5:

```
$ vcgencmd get_throttled  
0x0 → no throttling (good)  
0x50000 → under-voltage detected (bad PSU!)
```

Reading USB Charger Specs

You have two chargers with "USB 160W Max Power", but one is \$100+ and the other is <\$35.



The \$100 Spec:

5V=3A

9V=3A

15V=3A

20V=5A

28V=5A

28V=5.36A (150W Max)

Total Output: 160W Max

Single Port Maximum Output: 140W

Two Ports in Use: 160W Max

Three Ports in Use: 160W Max

Meaning of This Spec

Single Port

- Theoretically maximum $28V \times 5.36A \approx 150W$
- USB-PD 3.1 EPR support
- Requires 5A e-marked cable, so the practical real max is 140W ($28V \times 5A$)

Multi-Port

- **Total combined output = 160W**
- Not 160W per port
- Shared power budget

Possible distributions:

- 150W + 10W, 100W + 60W, or 65W + 65W + 30W

Total never exceeds:

160W

This is real total capacity — not marketing math.

The \$30 Spec:

USB-C1/C2 Output: 5V/3A, 9V/3A, 12V/3A, 15V/3A, 20V/3.25A, 65W Max

USB-A Output: 5V/3A, 9V/3A, 12V/2.5A, 15V/2A, 30W Max

Fast Charging Protocols: PD3.0/PD2.0/PD1.0 QC3.0/ QC2.0/QC1.0 PPS

Meaning of This Spec

Compared to the \$100 charger, this charger does not specify PD 3.1 EPR support (28V), so it is limited to $20V/3.25A = 65W$ max per port.

- Each port can do 65W individually
- But not at the same time

Marketing shows:

$$65W + 65W + 30W$$

But in reality: $65W + 30W = 95W$ total max

Why the Big Price Difference?

PD 3.1 EPR Support (28V)

True 28V output requires:

- Advanced PD controller
- Higher voltage MOSFETs
- Better isolation
- Thermal headroom

Cheap units may:

- Throttle under load
- Fake (or confusing) PDO tables

Internal Power Design0

A real 160W GaN charger needs:

- High-quality GaN transistors
- Strong PFC stage
- Large capacitors
- Proper heat sinking
- Protection (OVP, OCP, OTP, SCP)

These cost money.

Safety Certification

Premium brands pay for:

- UL / ETL testing
- FCC / CE compliance
- Thermal validation
- Liability insurance

Cheap brands may skip or minimize this.

Sustained vs Peak Power

Cheap 160W	Premium 160W
160W peak only	160W sustained
Likely throttling	Stable output
Voltage droop	Tight regulation
Unknown protection	Full protection